

Experiment 14

Understand feature scaling.

Program:

```
import numpy as np
import pandas as pd

df=pd.read_csv('/content/pre-process_datasample.csv')
df

df.Country.fillna(df.Country.mode()[0],inplace=True)

features=df.iloc[:, :-1].values
label=df.iloc[:, -1].values

from sklearn.impute import SimpleImputer

age=SimpleImputer(strategy="mean",missing_values=np.nan)
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)

age.fit(features[:, [1]])
Salary.fit(features[:, [2]])

features[:, [1]]=age.transform(features[:, [1]])
features[:, [2]]=Salary.transform(features[:, [2]])

from sklearn.preprocessing import OneHotEncoder
oh=OneHotEncoder(sparse_output=False)
Country=oh.fit_transform(features[:, [0]])

final_set=np.concatenate((Country,features[:, [1,2]]),axis=1)

from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)

from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)

feat_standard_scaler
feat_minmax_scaler
```

Output

```
StandardScaler output:
[[ 1.00000000e+00 -5.00000000e-01 -6.54653671e-01
   7.58874362e-01  7.49473254e-01]
 [-1.00000000e+00 -5.00000000e-01  1.52752523e+00
  -1.71150388e+00 -1.43817841e+00]
 [-1.00000000e+00  2.00000000e+00 -6.54653671e-01
  -1.27555478e+00 -8.91265492e-01]
 [-1.00000000e+00 -5.00000000e-01  1.52752523e+00
  -1.13023841e-01 -2.53200424e-01]
 [-1.00000000e+00  2.00000000e+00 -6.54653671e-01
   1.77608893e-01  6.63219199e-16]
 ...]

MinMaxScaler output:
[[1.         0.         0.         0.73913043  0.68571429]
 [0.         0.         1.         0.         0.         ]
 [0.         1.         0.         0.13043478  0.17142857]
 [0.         0.         1.         0.47826087  0.37142857]
 [0.         1.         0.         0.56521739  0.45079365]
 ...]
```